9-12 V DC INPUT

5 V LINEAR SUPPLY

J1
POWER IN

F1
T500mA / 250V

D1
1N5819

U1
LM7805CT / TO-220

JP1
MAIN PWR / closed

VIN_9_12
FUSED_VIN
REG_IN
REG_5V
+5V
GND

C2
47uF / 25V

C1
330nF / 25V

C3
100nF / 16V

C4
100uF / 16V

JP2
HAPTIC PWR / closed

HAPTIC_5V

HAPTIC DRIVER

U4
MAR / Adafruit 2305 QT

VIN
SCL
SDA
IN/TRIG
GND
OUT+
OUT-
MOT_P
MOT_N
HAP_TRIG

M1
3V ERM / Adafruit 1201

J6
MOTOR

MOTOR
(off board)

UART TO NEXT BOARD

J3
DOWNSTREAM / IDC 2x4

UART_TX
GND

J2
UPSTREAM / IDC 2x4

UART_RX
GND

UART FROM KHUN

PIC18F57Q43 CURIOSITY NANO

U2
MAR / PIC18F57Q43 Curiosity Nano

NC
ID
CDC_RX
CDC_TX
RB6/DBG1
RB4/DBG2
RC2/TX
RC3/RX
RB2/SDA
RB1/SCL
RC4
RC5
RC6
RD4
GND
RA3
RA4
RB2
RC7
RD0
RD1
RD2
RD3
GND
RF0
RF1
RF2
RF3
VBUS
VOFF
RE3/MCLR
RB7/DBG0
GND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0

I2C_SDA
I2C_SCL
EXP_RC4
AMP_SD
AUDIO_PWM
EXP_RA2
EXP_RA1
EXP_RA0
DBG_LED
DBG_BTN

R8
10k / 0.25W

C5
100nF / 16V

C6
10uF / 16V

C11
100nF / 16V

C12
10uF / 16V

AUDIO FILTER + SPEAKER DRIVER

U3
MAR / Adafruit 2130

A+
A-
SD
VIN
GND
OUT+
OUT-
SPK_P
SPK_N

R4
22k / 0.25W

R6
1k / 0.25W

R5
1k / 0.25W

C8
47nF / 16V

C9
47nF / 16V

R7
470R / 0.25W

LS1
8 ohm / 1W / 3923

J5
SPEAKER

SPEAKER
(off board)

C10
100nF / 16V

EXPANSION

J4
SPARE HEADER

EXP_RA0
EXP_RA1
EXP_RA2
EXP_RC4
+5V
GND

R1
1k / 0.25W

LED_A

D2
GREEN

R2
10k / 0.25W

DEBUG LED

C7
1uF / 16V

R3
1k / 0.25W

BTN_SW

DEBUG BUTTON

SW1
DEBUG / N.O.

MAR = Mohammed Al Rasbi
U2 is the complete Nano board.
VOFF is grounded; VBUS is unused.
U3/U4 are Adafruit modules.
Use QT revision of Adafruit 2305.
All resistors: 0.25 W.
All signal capacitors: >=16 V.

TEST POINTS

TP1 +5V
TP2 GND
TP3 UART_RX
TP4 UART_TX
TP5 I2C_SCL
TP6 I2C_SDA
TP7 AUD_IN

U1: heatsink <=15 C/W; allow ~2.5 W at 12 V / 0.35 A.
U4: I2C 0x5A, ERM mode, nominal 3 V motor setting.
U3: start trim at minimum; speaker outputs must float.
Confirm team UART levels and run KiCad ERC before submission.

File: Mohammed-Interpretation.kicad_sch

Interpretation Board - Mohammed Al Rasbi

Size: A2 Date: 2026-10-23 Rev: C Sheet: 1/1